

ON PHYLOGENETIC RANKS OF GRAPHS

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ABSTRACT. In this paper we consider the problem of embedding metric graphs into a product of trees. The minimum number of trees to this is called the *phylogenetic rank* of the graph. We construct an algorithm for determining this value and investigate this value for many classes of graphs including small graphs, random graphs.

OVERVIEW AND TODOS

This is a temporary section for what's last changed in the file and what is left to be done.

Updates.

- Added Theorem 5.1: Graph theoretic characterisation of rank one graphs
- Added proof Proposition 2.8: cycle graphs have rank $\lceil n/2 \rceil$
- Structured paper with Introduction, Background, Algorithm, Experimental / Families of Graphs, and Conjectures sections.
- Added background on some graph notation, added definitions for phylogenetic / tree rank (I noticed that Speyer-Sturmfels expository article seems to be based on the Pachter-Sturmfels computational biology book (which is in sources folder), which uses the term **tree rank**) - but now, having looked through more literature, tree rank is confusing as it appears in different places.
- I like the idea in Remark 2.10, should I make this more rigorous?

Todos.

- Add algorithms
- Add proofs of algorithms
- Find source for 4-point condition
- Add results from experiments
- Set up terminology for random graphs

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Key words and phrases. foo.

1. INTRODUCTION

2. BACKGROUND

Notation for graphs. In this paper, a graph $G = (V, E)$ consists of a finite set V of vertices and set $E \subseteq \binom{V}{2}$ of edges. In particular, the graph G has no loops or multi-edges. An *edge-length map* $\ell : E \rightarrow \mathbb{R}_{>0}$ assigns a length to each edge of G . A *walk* in G is a sequence $w = (v_0, e_1, v_1, e_2, v_2, \dots, v_{m-1}, e_m, v_m)$ such that $e_i = \{v_{i-1}, v_i\} \in E$ for each $i \in [m]$. A *path* is a walk that does not repeat vertices or edges. The walk w is a *cycle* if $v_m = v_0$ and all other edges and vertices are distinct. A graph is *connected* if for any pair of vertices there is a path joining them. A *tree* is a connected graph that contains no cycles.

Fix a graph G and edge-length map ℓ for G . The map ℓ naturally gives a notion of length to any walk $w = (v_0, e_1, \dots, e_m, v_m)$ by $\ell(w) := \ell(e_1) + \ell(e_2) + \dots + \ell(e_m)$. The *induced path metric* on G is a metric on V is given by

$$d : V \times V \rightarrow \mathbb{R}_{\geq 0} \cup \{\infty\}, \quad d(p, q) := \min\{\ell(w) : w = (p, \dots, q) \text{ path}\},$$

where, by convention, we take $\min(\emptyset) = \infty$ if there are no paths from p to q . Note that, since the graph is finite, there are finitely many paths from p to q .

Definition 2.1 (Metric graph) A *metric graph* is a pair (G, d) where $G = (V, E)$ is a graph and $d : V \times V \rightarrow \mathbb{R}_{\geq 0}$ is a metric on V that is the induced path metric of some edge-length map on E . We treat (G, d) as a finite metric space with underlying set V , and by a common abuse of notation we write G for (G, d) and disambiguate metrics by writing d_G for the metric on G .

Definition 2.2 (Tree embedding) Let G be a metric graph. A *tree embedding* for G is injective continuous map of metric spaces $G \rightarrow T_1 \times T_2 \times \dots \times T_k$ where: T_i is a metric graph whose underlying graph is a tree; and the product space is imbued with the supremum-metric. Explicitly, given two points $p, q \in T_1 \times \dots \times T_k$, the distance between them is $d(p, q) := \max\{d_{T_i}(p_i, q_i) : i \in [k]\}$.

We say that a tree embedding $\iota : G \rightarrow T_1 \times \dots \times T_k$ is *minimal* if for each $i \in [k]$, the restriction of ι to $T_1 \times \dots \times \widehat{T_i} \times \dots \times T_k$ is not injective. **Franklin: Should this be "not isometry"** **Jackson: test**

We note that the notion of tree embeddings applies to any metric spaces. However, for our purposes, we will keep to embedding graphs.

Remark 2.3 In the definition of a tree embedding, we put the supremum-metric on the product of metric trees. In [PS05, Section 3.5] this construction is called a *mixture of metrics*, which is defined with the purpose of studying comparative genomics. In that book, these mixtures are used to study the tropicalisation of Pfaffian varieties, which are geometrically analogous to secants varieties.

Definition 2.4 (Phylogenetic / tree rank) Let G be a metric graph. The *phylogenetic rank* or *tree rank* of G is the smallest positive integer r_G such that there exists a tree embedding $G \rightarrow T_1 \times \cdots \times T_{r_G}$.

Note that the empty product of metric spaces is the empty space. Thus a graph has phylogenetic rank zero if and only if $V = \emptyset$, i.e., the graph G is the totally empty graph.

Example 2.5 Some easy examples of phylogenetic rank:

- If T is a metric graph whose underlying graph is a tree then $r_T = 1$ since $T \rightarrow T$ is an embedding,
- The complete graph $K_n = ([n], \binom{[n]}{2})$ whose edges all have the same length $\ell(e) = u \in \mathbb{R}_{>0}$ has phylogenetic rank one as there is an embedding $K_n \rightarrow T$ where: $T = (\{0, 1, \dots, n\}, \{\{0, i\} : i \in [n]\})$ is the star graph whose edges all have length $\ell(\{0, i\}) = u/2$; and the embedding map sends vertex $i \in V(K_n)$ to $i \in V(T)$.
- If G is not connected, then the phylogenetic rank is infinite.

Phylogenetic rank one spaces are characterised by the *four point condition*.

Proposition 2.6 ([Bun74, Theorem 2], [PS05, Theorem 2.35]) *A metric graph $G = (V, E)$ has phylogenetic rank one if and only if it satisfies the **four point condition**: for any distinct $s, t, u, v \in V$ the maximum value among*

$$d(s, t) + d(u, v), \quad d(s, u) + d(t, v), \quad d(s, v) + d(u, v)$$

is attained at least twice.

The smallest example of a graph that does not have phylogenetic rank one is $K_4 \setminus e$.

Example 2.7 Let G be the metric graph whose underlying graph is $K_4 \setminus e$ and whose edges all have length one. Explicitly, let us write

$$V(G) = [4], \quad E(G) = \{12, 13, 23, 14, 24\}.$$

To see that the metric does not satisfy the four point condition, observe that

$$3 = d(1, 2) + d(3, 4) > d(1, 3) + d(2, 4) = d(1, 4) + d(2, 3) = 2.$$

Therefore, the metric graph G must have phylogenetic rank at least two. Giving a construction to show that the phylogenetic rank is exactly two is straightforward.

The first non-trivial family of examples are the cycle graphs.

Proposition 2.8 *Let C_n be the metric graph consisting on an n -cycle whose edges all have length one. Then the phylogenetic rank of C_n is $\lceil n/2 \rceil$. If G is a metric graph whose underlying graph is an n -cycle with another metric, then the phylogenetic rank is at most $\lceil n/2 \rceil$.*

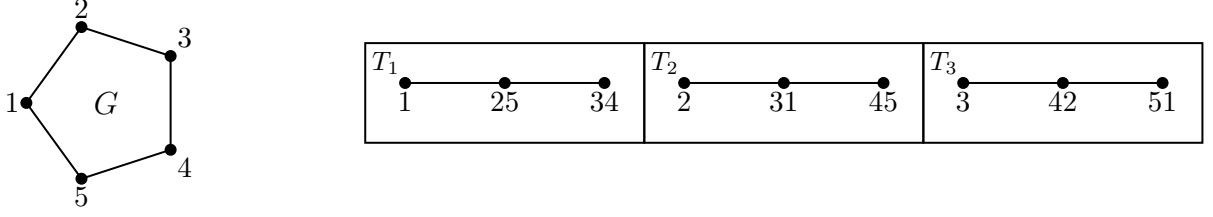


FIGURE 1. Example of a unit-edge-length cycle embedded into a product of three trees in the proof of Proposition 2.8. The vertices of each tree T_i are v_0, v_1, v_2 from left to right.

Proof. Throughout the proof, we write $m := \lceil n/2 \rceil$. First we give an embedding to show that $r_{C_n} \leq m$. Let $V(C_n) = [n]$ and assume that the vertices are labelled cyclically so that the edges are $\{1, 2\}, \{2, 3\}, \dots, \{n-1, n\}, \{1, n\}$. Let T be the metric path graph with m vertices and whose edges have unit length. We label the vertices $V(T) = \{v_0, v_1, \dots, v_{m-1}\}$. We now give an embedding of C_n into T^m using the notation: for each $x \in \mathbb{Z}$, we write $\bar{x}$ for its remainder on dividing by n . We define the map

$$\iota : C_n \rightarrow T^m, \quad (\iota(i))_j = v_k \text{ where } k = \begin{cases} \bar{i-j} & \text{if } \bar{i-j} \leq n/2, \\ n - (\bar{i-j}) & \text{if } \bar{i-j} > n/2, \end{cases}$$

for each $i \in V(C_n) = [n]$ and $j \in [m]$. It is relatively straightforward to see that ι is a tree embedding. So $r_{C_n} \leq \lceil n/2 \rceil$. Similarly, we note that if the graph C_n has a different metric, then a similar construction gives the same upper bound.

Now suppose that C_n is a unit edge-length metric graph embedded into a product of strictly fewer than m trees, say $\iota : C_n \rightarrow T_1 \times \dots \times T_k$ for some $k < m$. We take cases on the parity of n .

Assume that n is even. For each $i \in [m]$ note that the distance, in C_n , between the vertices i and $i+m$ is $d_{C_n}(i, i+m) = m$. So, for each $i \in [m]$ there is some coordinate $j = j_i \in [k]$ of the product space such that $d_{T_j}(\iota(i)_j, \iota(i+m)_j) = m$. Since $k < m$, by pigeonhole principal, there exists distinct $i, i' \in [m]$ such that $j_i = j_{i'} =: j$. Since $i \neq i'$ and $i \neq i' + m$, it follows that

$$d_{C_n}(i, i') < m, d_{C_n}(i+m, i'+m) < m, d_{C_n}(i, i'+m) < m, d_{C_n}(i', i+m) < m.$$

Since $d_{C_n}(i, i+m) = d_{C_n}(i', i'+m) = d_{T_j}(\iota(i)_j, \iota(i+m)_j) = d_{T_j}(\iota(i')_j, \iota(i'+m)_j) = m$, by the four point condition, we have that either

$$\begin{aligned} d_{T_j}(\iota(i)_j, \iota(i')_j) + d_{T_j}(\iota(i+m)_j, \iota(i'+m)_j) &= 2m \\ \text{or } d_{T_j}(\iota(i)_j, \iota(i'+m)_j) + d_{T_j}(\iota(i')_j, \iota(i+m)_j) &= 2m. \end{aligned}$$

But in either case, this contradicts the lengths of the edges in C_n .

A similar proof holds when n is odd. □

Proposition 2.9 *Let G be a metric graph with n vertices, then the phylogenetic rank is at most $n - 1$.*

Proof. We write $G = (V, E)$. For each $v \in V$, we construct a metric tree T_v as follows whose underlying graph is a path. The vertex set of T_v is given by $V(T_v) := \{d_G(v, w) : w \in V\} \subset \mathbb{R}_{\geq 0}$ from which we directly define the metric on T_v by $d_{T_v}(a, b) = |a - b|$. It is straightforward to check that this is indeed the metric graph whose underlying graph is a path and whose edges are the pairs $\{a, b\} \in \binom{V(T_v)}{2}$ such that there exists no $c \in V(T_v)$ that lies strictly between a and b .

Let V' be the set V with one vertex removed. We define the map $\iota : G \rightarrow \prod_{v \in V'} T_v$ by $(\iota(x))_v = d_G(v, x) \in V(T_v)$. Let us check that this is a tree embedding. Fix a pair of distinct vertices $p, q \in V$. Without loss of generality, we may assume $p \in V'$. For each $v \in V'$, by the reverse triangle inequality, we have

$$d_{T_v}(\iota(p)_v, \iota(q)_v) = |d_G(v, p) - d_G(v, q)| \leq d_G(p, q).$$

So $d(\iota(p), \iota(q)) \leq d_G(p, q)$. Since $p \in V'$, the p -th coordinate of the product gives us $d_{T_p}(\iota(p)_p, \iota(q)_p) = |d_G(p, p) - d_G(p, q)| = d_G(p, q)$. Thus $d(\iota(p), \iota(q)) = d_G(p, q)$. $\square$

Remark 2.10 It is possible to give an essentially one-line Tropical proof of Proposition 2.9 by exploiting connection between metric trees, Tropical Grassmannians, secant varieties, and tree-embeddings. The proof is as follows. Since the $(n - 1)$ -th secant variety of Grassmannian $\text{Gr}(2, n)$ under the Plücker embedding is dense, tropicalising this tells us that the $(n - 1)$ -fold product of tree metrics can realise any metric. [Maybe this argument also tells us about other metrics other than those coming from graphs.](#) Perhaps our conjecture that $r_G \leq \lceil n/2 \rceil$ is telling us that [graph metrics are special points in the space of all metrics.](#)

For each graph G there is a ‘natural’ metric graph whose edges have unit length. Identifying graphs with metric graphs in this way, we see that for each fixed positive integer r , the set of graphs with phylogenetic rank at most r forms a *hereditary class* of graphs.

Proposition 2.11 *Fix a positive integer r . The set of graphs G such that the corresponding unit edge-length metric graph has phylogenetic rank at most r , forms a hereditary class, i.e., the set of graphs is closed under taking induced subgraphs. Moreover, for this hereditary class, the minimal forbidden induced subgraphs include the cycles C_i for $i \geq \max\{4, 2r + 1\}$.*

Proof. The restriction of a tree embedding to an induced subgraph is a tree embedding, so the phylogenetic rank of the induced subgraph is bounded above the phylogenetic rank of the original graph. By Proposition 2.8, the cycle C_i has phylogenetic rank $\lceil i/2 \rceil$ if $i \geq 4$ and C_3 has phylogenetic rank one because it is a complete graph. Since all proper connected induced subgraphs of the cycle are trees, the cycles are minimal forbidden induced subgraphs if $i \geq \max\{4, 2r + 1\}$. $\square$

3. HEURISTIC ALGORITHM

Our first main result gives an algorithm for computing the phylogenetic rank of a metric graph. For the purposes of writing an algorithm, we use the *distance matrix* D of a metric graph (G, d) . The distance matrix is $D \in \mathbb{R}^{V \times V}$ where $D_{x,y} = d(x, y)$.

Note:

- If G is a metric graph, then their metric is denoted by d_G .
- We try to keep the embedding dimension low.
- We construct the image trees vertex by vertex.

JULIO: WE NEED TO CHOOSE NOT ONLY ANY EDGE, BUT THE EDGE THAT GIVES US THE BEST SCORE

Algorithm 3.1

Input: G , a connected simple metric graph with vertex set $V = \{v_1, \dots, v_n\}$.

Output: $\iota: G \rightarrow \Gamma_1 \times \dots \times \Gamma_r$, an isometry where

- (1) $\Gamma_1, \dots, \Gamma_r$ are metric trees, and
 - (2) $\Gamma_1 \times \dots \times \Gamma_r$ is equipped with the supremum norm.
- 1: $D := (d_G(v_i, v_j))_{v_i, v_j \in V} \in \mathbb{R}_{\geq 0}^{V \times V}$.
 - 2: $r := 0$.
 - 3: **while** $D \neq 0$ **do**
 - 4: $((v_i, v_j), \mathcal{I}) := \text{PickPairAndCandidates}(D, \{G \xrightarrow{\iota_i} \Gamma_i \mid i \in [r]\})$.
 - 5: **for** $(G \xrightarrow{\iota_k} \Gamma_k) \in \mathcal{I}$ and $e_k \in E(\Gamma_k)$ **do**
 - 6: $(A, b) := \text{ConstructConstraints}(G \xrightarrow{\iota_k} \Gamma_k, v_i, v_j, e_k)$
 - 7: $(p, x) := \text{CanSolveAndPickSolution}((A, b), D)$
 - 8: **if** $p == \text{true}$ **then**
 - 9: $(G \xrightarrow{\iota_k} \Gamma_k, D) := \text{ExtendAndUpdate}(G \xrightarrow{\iota_k} \Gamma_k, D, e_k, x)$.
 - 10: **goto** Step 3.
 - 11: $(G \xrightarrow{\iota_{r+1}} \Gamma_{r+1}, D) := \text{InitializeAndUpdate}((G, v_i, v_j), D)$
 - 12: $r := r + 1$.
 - 13: **for** $i \in [r]$ **do**
 - 14: $G \xrightarrow{\iota_i} \Gamma_i := \text{Totalize}(G \xrightarrow{\iota_i} \Gamma_i)$
 - 15: **return** $G \xrightarrow{\iota_1 \times \dots \times \iota_r} \Gamma_1 \times \dots \times \Gamma_r$.

Quick notes (each function to me made its own algorithm):

PickPairAndCandidates: pick a pair of vertices to be “worked on” and candidates:

- For pair, we recommend $(v_i, v_j) = \text{argmax}(D)$, as we believe it that realizing it has the highest chance for realizing smaller distances as a knock-on effect. Note that zeroes in D are either diagonal entries or already realized distances, so $d_G(v_i, v_j)$ is the largest unrealized distance. If argmax is not unique, employ a sensible tiebreaker.

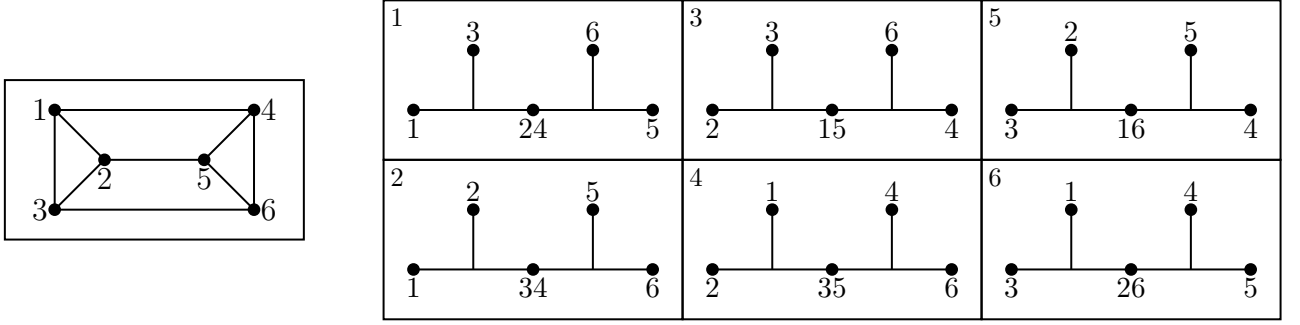


FIGURE 2. Left: prism graph G on $[6]$ with unit edge lengths. Right: six metric tree images of G , in each tree the edges have length $1/2$.

Franklin: I think the embedding is wrong, 3 and 6 are 2 apart. Ollie: Yes, this isn't clear, the edges on the right-hand graphs have length $1/2$ so in each tree they are distance one apart - I added clarification above; let me know if there's still an issue.

- The candidates are all partial maps $G \xrightarrow{\iota_k} \Gamma_k$ whose codomain contains exactly one of v_i or v_j . This means we can potentially realize $d_G(v_i, v_j)$ by extending Γ_k by a vertex w and mapping v_i or v_j to it.

ConstructConstraints: construct the constraints for $d_G(v_i, v_j)$ to be realized by attaching a new vertex w to the edge e_k . The three variables are distance between w and the attaching point on e as well as the two distances between the attaching point on e and the end-vertices of e .

ExtendAndUpdate: Extends Γ_k as described above and update D by setting all newly realized distances to 0.

InitializeAndUpdate: Create a new $G \xrightarrow{\iota_{r+1}} \Gamma_{r+1}$ realizing $d_G(v_i, v_j)$ and update D by setting all newly realized distances to 0.

Totalize: extend the 1-Lipschitz partial map $G \xrightarrow{\iota_i} \Gamma_i$ to a 1-Lipschitz total map $G \xrightarrow{\iota_i} \Gamma_i$.

Theorem 3.2 *Algorithm ?? terminates and correctly returns the set of minimal tree embeddings.*

Example 3.3 Let G be the prism graph with unit edge lengths, see Figure 2.

4. HOLLISTIC ALGORITHM

Definition 4.1 Let G be a simple connected metric graph with vertex set V and edge set E .

- (1) A finite sequence of adjacent vertices $\gamma = (v_0, \dots, v_k) \in V^k$ is a *geodesic* or *length minimizing path* of G if $\sum_{i=l_1}^{l_2} d_G(v_i, v_{i+1}) = d_G(v_{l_1}, v_{l_2})$ for all $0 \leq l_1 < l_2 \leq k$.

- (2) Let $\mathcal{G}$ denote the set of all maximal geodesics, where $\gamma_1 \leq \gamma_2$ if and only if γ_1 is a contiguous subsequence of γ_2 .
- (3) A finite set of geodesics $C \subseteq \mathcal{G}$ is *(tree-)compatible*, if there exists a metric tree Γ_C and a 1-Lipschitz map $\iota_C: G \hookrightarrow \Gamma_C$ such that $\iota_C|_\gamma$ is an isometry for all $\gamma \in C$. Here, $\iota_C|_\gamma$ refers to ι_C restricted to the edges connecting subsequent vertices in γ .
- (4) Let $\mathcal{C} \subseteq 2^{\mathcal{G}}$ denote the set of maximal compatible subsets, where $C_1 \leq C_2$ if and only if there is a $\varphi: C_1 \rightarrow C_2$ such that $\gamma \subseteq \varphi(\gamma)$ for all $\gamma \in C_1$.
- (5) The *distance realization map* is defined as

$$\pi: \mathcal{C} \longrightarrow 2^{\binom{V}{2}}, \quad C \longmapsto \left\{ \{ij\} \in \binom{V}{2} \mid d_{\Gamma_C}(i, j) = d_G(i, j) \right\}.$$

Lemma 4.2 *Let $\iota = (\iota_1, \dots, \iota_r): G \rightarrow \Gamma_1 \times \dots \times \Gamma_r$ be a minimal embedding. Then for any maximal geodesic γ , there is some Γ_i such that $\iota_i|_\gamma$ is an isometry.*

Proof. **Yue:** **TODO: insert proof here.** □

Algorithm 4.3 (All embeddings)

Input: G , a simple connected metric graph

Output: a set of embeddings that contains all minimal embeddings

- 1: Construct $\mathcal{G}$.
- 2: Construct $\mathcal{C}$.
- 3: Construct $\pi: \mathcal{C} \rightarrow 2^{\binom{V}{2}}$.
- 4: Find all minimal subsets $\{C_1, \dots, C_r\} \subseteq \mathcal{C}$ such that $E = \bigcup_{i=1}^r \pi(C_i)$.
- 5: **return** those subsets

Proof. **Yue:** **TODO: flesh out this proof!**

- (1) Let $\iota: G \rightarrow \Gamma_1 \times \dots \times \Gamma_r$ be a minimal embedding.
- (2) Use Lemma 4.2, to construct $C_1, \dots, C_r \in \mathcal{C}$ such that it gives rise to ι (take C_i to be the set of maximal geodesics on whom ι_i is an isometry).
- (3) The set $\{C_1, \dots, C_r\}$ is minimal because ι was minimal. □

Things to do for implementation:

- Implement (and write down) algorithm for checking whether $C \subseteq \mathcal{G}$ is compatible via 4-point condition and feasibility of tropical polyhedra.
- Implement a general framework for poset traversal and use it to compute all maximal geodesics of a graph.

More things to do:

- Add an example (compute $\mathcal{G}$, $\mathcal{C}$, the possible $C_1, \dots, C_r$, as well as the posets that give rise to them).

5. FAMILIES OF GRAPHS

5.1. Phylogenetic rank one. In this small section, we classify graphs with phylogenetic rank one. The main result is Theorem 5.1, which intuitively says: firstly that the all phylogenetic rank one graphs are “trees of cliques” (or more concretely, they are 1-sums of cliques); and secondly the only obstructions to phylogenetic rank one are cycles of length four or more, and $K_4 \setminus e$, see Proposition 2.8 and Example 2.7. We now recall all the necessary terminology for the result.

Let G be a graph and C a cycle of G . A *chord* of C is an edge of G that connects two non-consecutive vertices of C . We say that a graph G is *chordal* if every cycle of length at least four has a chord. Let $H = (H_1, H_2, \dots)$ be a (possibly infinite) sequence of graphs. We say that G is H -free if no induced subgraph of G is isomorphic to any H_i .

Given two graphs G and H (on potentially different vertex sets), their 1-sum is the graph obtained from their disjoint union with one vertex in each of G and H identified. We say that G is a *1-sum of cliques* if it can be obtained from a clique by performing a sequence of 1-sums with cliques. For instance, a tree is 1-sum of cliques as it can be obtained from a single vertex by adding one degree-one vertex at a time. Note that if G is a 1-sum of cliques, then the decomposition of G into maximal cliques is unique and is given by the set of maximal cliques of G .

Theorem 5.1 *Let G be a metric graph with unit edge lengths. Then the following are equivalent:*

- (1) G has phylogenetic rank one,
- (2) G is free of $K_4 \setminus e$ and the cycle graphs C_i for all $i \geq 4$,
- (3) G is chordal and $(K_4 \setminus e)$ -free,
- (4) G is a 1-sum of cliques.

Proof. (1) $\implies$ (2). Assume that G has phylogenetic rank one, hence there is an embedding of G into a metric tree T . By Example 2.7 and Proposition 2.8, it follows that G cannot contain an induced copy of $K_4 \setminus e$ or a cycle C_i for $i \geq 4$ because the tree embedding could be restricted to that subgraph.

(2) $\implies$ (3). This implication follows immediately from the definition.

(3) $\implies$ (4). Assume that G is chordal and contains no copies of $K_4 \setminus e$. We will use the well-known characterisation of chordal graphs in terms of *perfect elimination orderings*, which says that there is an ordering of the vertices $v_1 < v_2 < \dots < v_n$ of G such that for each $i \in [n]$ the vertices greater than v_i in its neighbourhood form a clique, explicitly the set $\{v_j \in V(G) : v_i < v_j, v_i v_j \in E(G)\}$ is a clique of G . For ease of notation let us write $N_G(v) = \{w \in V(G) : vw \in E(G)\}$ for the neighbourhood of v in G , and note that $v \notin N_G(v)$.

To show that G is a 1-sum of cliques, we proceed by induction on n the number of vertices of G . The base case is the graph with one vertex, which is trivially a 1-sum

of cliques. For the induction step, let us assume that we have a perfect elimination order $v_1 < \dots < v_{n-1} < v_n$ and that there is a 1-sum decomposition of $G[V']$ into cliques where $V' = \{v_2, \dots, v_n\}$. We now check that G has a 1-sum decomposition into cliques by taking cases on the size of neighbourhood $N_G(v)$ of v .

Case 1. Assume $|N_G(v)| = 1$, then G is a 1-sum of $G[V']$ and K_2 , and we are done.

Case 2. Assume that $|N_G(v)| \geq 2$. Then, by the perfect elimination ordering, we have that $N_G(v)$ is a clique of $G[V']$. Next we show that $N_G(v)$ must be a clique of the 1-sum decomposition of $G[V']$, i.e., a maximal clique of $G[V']$. Since $N_G(v)$ is a clique, it must be contained in a maximal clique of $G[V']$. If $N_G(v)$ is not maximal, then take x be any vertex of that maximal clique that is not in $N_G(v)$ and let $y, z \in N_G(v)$ be any two distinct vertices. Then the set $\{v, x, y, z\}$ forms an induced $K_4 \setminus e$, a contradiction. Thus $N_G(v)$ is a maximal clique of G . So, by adding v to the clique $N_G(v)$ in the 1-sum decomposition of $G[V']$, we obtain a 1-sum decomposition of G into cliques.

(4) $\implies$ (1). Assume that G is a 1-sum of cliques. We will show that G embeds into a metric tree by induction on the number of cliques. If G has one clique, i.e., $G = K_n$, then it embeds into the star graph, see Example 2.5. For the induction step, assume that G has phylogenetic rank one and embeds into the tree T . Let $G' = K_m$ be a clique, which, by the above, embeds into a metric tree T' . It is easy to see that a 1-sum of G and G' embeds, naturally, into a 1-sum of T and T' . However, a 1-sum of trees T and T' is a tree, thus a 1-sum of G and G' has phylogenetic rank one. This completes the proof. $\square$

5.2. Families with phylogenetic rank $n/2$. We have already seen in Proposition 2.8 that the n -cycle has phylogenetic rank $\lceil n/2 \rceil$. We will exhibit some other families of graphs that also obtain this conjectured upper bound.

The hypercube Q_n is the convex hull of the 0/1-lattice points in $\mathbb{R}^n$. We identify Q_n with its vertex-edge graph which has 2^n vertices, the length n binary strings $\{0, 1\}^n$, where two strings are adjacent if and only if they differ in exactly one position.

The (k, n) -hypersimplex is the polytope obtained by intersecting the hypercube Q_n with the affine hyperplane of points $\{x \in \mathbb{R}^n : x_1 + \dots + x_n = k\}$ whose coordinate sum is k . Its vertices are the length n binary strings with exactly k ones and its edges are xy where $x - y$ is a vector with exactly one 1 and one -1 .

Theorem 5.2 *For $n \geq 5$, the following graphs have phylogenetic rank $n/2$:*

- $K_n \setminus C_n$,
- K_n minus a perfect matching (where n is even),
- The vertex-edge graph of the hypercube Q_k on $n = 2^k$ vertices,
- K_n minus two edge-disjoint cycles (where $n \geq 8$),
- The vertex-edge graph of the $(2k, k)$ -hypersimplex on $n = \binom{2k}{k}$ vertices

Proof. For each graph we can partition the vertices into pairs such that the distance between the pair is equal to the diameter of the graph. For instance the diameter of K_n minus a perfect matching is 2, and so the perfect matching gives us the partition. For each pair of pairs, there is no way to simultaneously isometrically map both pairs into a tree while not increasing any of the other distances. Thus any tree embedding must contain at least $n/2$ trees. To see that $n/2$ is possible, it is straightforward to give a direct construction. $\square$

5.3. Random graphs.

6. CONJECTURES

One motivating problem from computation Biology comes from the end of Chapter 4 from [PS05], which generalises the four point condition. Below we recall this problem in the language of metric spaces. *In the book, they write this in terms of mixtures of metrics. I think the following is equivalent to this conjecture without the alg-stats language. I've uploaded the book, see './sources/pachter-algebraic.pdf', the relevant part is on page 132 of the book (page 140 of pdf).*

Conjecture 6.1 Let (X, d) be a finite metric space and fix a positive integer r . Suppose that for each collection of $2r + 2$ points $U \subseteq X$, the phylogenetic rank of $(U, d|_U)$ is at most r , then the phylogenetic rank of X is at most r .

Maybe we can show this for metric graphs? Or, say, in the case $r = 2$?

One can generalise the four point condition in a natural way. Given a metric on a finite space X , we say that X satisfies the $2r + 2$ point condition (for positive integer r) if for any $(2r + 2)$ -subset $Y \subseteq X$, the maximum

$$\max \left\{ \sum_{pq \in M} d(p, q) : M \text{ perfect matching of complete graph on } Y \right\}$$

is attained at least twice. It is known that if X has phylogenetic rank at most r then it satisfies the $2r + 2$ point condition [PS05, Proposition 3.49]. Moreover, the converse does not hold, see [PS05, Example 3.51], which is a copy of K_6 minus a cycle whose edges have unit length. Explicitly, this is a metric graph that satisfies the 6-point condition but embeds minimally into a product of three trees.

Example 6.2 Figure 3 shows an example of a minimal embedding of G , the graph K_6 minus a 6-cycle, into a product of three trees.

Among graphs with a fixed number of vertices, we expect that the cycle with unit edge lengths has the largest phylogenetic rank.

Conjecture 6.3 Let G be a graph with n vertices then $r_G \leq \lceil n/2 \rceil$.

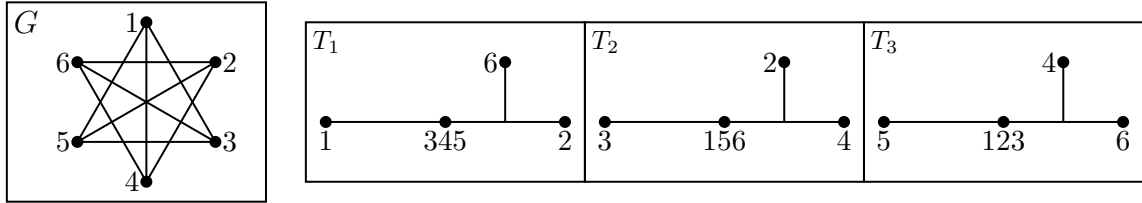


FIGURE 3. A minimal embedding of K_6 minus a cycle into three trees.

Example 6.4 All graphs on 5 vertices have phylogenetic rank at most 3. There is only one graph on 5 vertices with phylogenetic rank 3, which is the cycle C_5 .

Conjecture 6.5 The phylogenetic rank is lower-semicontinuous / upper-semicontinuous with respect to the metric.

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